

Machine Learning Fundamentals

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1. Introduction

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed.

2. Types of Machine Learning

- Supervised Learning: Uses labeled data. Examples: Linear Regression, Logistic Regression, Decision Trees.
- Unsupervised Learning: Uses unlabeled data. Examples: K-Means Clustering, PCA.
- Reinforcement Learning: Agents learn through rewards and penalties.

3. ML Workflow

1. Collect Data
2. Clean and Preprocess Data
3. Feature Engineering
4. Split Data into Training and Testing Sets
5. Train Model
6. Evaluate Model
7. Deploy Model
8. Monitor Performance

4. Important Concepts

- Features: Input variables.
- Target Variable: Output variable.
- Training Set: Data used for learning.
- Test Set: Data used for evaluation.
- Validation Set: Data used for tuning.

5. Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines
- K-Nearest Neighbors
- Neural Networks

6. Evaluation Metrics

Classification:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Regression:

- MAE
- MSE
- RMSE
- R-Squared

7. Overfitting and Underfitting

Overfitting occurs when a model memorizes training data and performs poorly on unseen data. Underfitting occurs when a model fails to capture patterns in the data.

8. Feature Engineering

Includes scaling, normalization, encoding categorical variables, and feature selection.

9. Deep Learning

A subset of ML that uses neural networks with multiple layers to learn complex patterns.

10. Applications

- Recommendation Systems
- Fraud Detection
- Healthcare Diagnostics
- Computer Vision
- Natural Language Processing
- Autonomous Vehicles

11. Conclusion

Machine Learning is a foundational technology for modern AI systems and is widely used across industries.